6 External conditions

6.1 General

The external conditions described in this clause shall be considered in the design of a wind turbine.

Wind turbines are subjected to environmental and electrical conditions that may affect their loading, durability and operation. To ensure the appropriate level of safety and reliability, environmental, electrical and soil parameters shall be taken into account in the design and shall be explicitly stated in the design documentation.

The environmental conditions are further divided into wind conditions and other environmental conditions. The electrical conditions refer to the electrical power network conditions. Soil properties are relevant to the design of wind turbine foundations.

The external conditions are subdivided into normal and extreme categories. The normal external conditions generally concern recurrent structural loading conditions, while the extreme external conditions represent rare external design conditions. The design load cases shall consist of potentially critical combinations of these external conditions with wind turbine operational modes and other design situations.

Wind conditions are the primary external conditions affecting structural integrity. Other environmental conditions also affect design features such as control system function, durability, corrosion, etc.

The normal and extreme conditions, which are to be considered for design according to wind turbine classes, are prescribed in the following subclauses.

6.2 Wind turbine classes

The external conditions to be considered for design are dependent on the intended site or site type for a wind turbine installation. Wind turbine classes are defined in terms of wind speed and turbulence parameters. The intention of the classes is to cover most applications. The values of wind speed and turbulence parameters are intended to represent many different sites and do not give a precise representation of any specific site, see 11.3. The wind turbine classification offers a range of robustness clearly defined in terms of the wind speed and turbulence parameters. Table 1 specifies the basic parameters, which define the wind turbine classes.

A further wind turbine class, class S, is defined for use when special wind or other external conditions or a special safety class, see 5.3, are required by the designer and/or the customer. The design values for the wind turbine class S shall be chosen by the designer and specified in the design documentation. For such special designs, the values chosen for the design conditions shall reflect an environment at least as severe as is anticipated for the use of the wind turbine.

The particular external conditions defined for classes I, II and III are neither intended to cover offshore conditions nor wind conditions experienced in tropical storms such as hurricanes, cyclones and typhoons. Such conditions may require wind turbine class S design.

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Table 1 – Basic parameters for wind turbine classes[1]

Wind turbine class		I	II	III	S
V_{ref}	(m/s)	50	42,5	37,5	Values
A	I_{ref} (-)	0,16			specified
B	I_{ref} (-)	0,14			by the
C	I_{ref} (-)	0,12			designer

In Table 1, the parameter values apply at hub height and

V_{ref} is the reference wind speed average over 10 min,

A designates the category for higher turbulence characteristics,

B designates the category for medium turbulence characteristics,

C designates the category for lower turbulence characteristics and

I_{ref} is the expected value of the turbulence intensity[2] at 15 m/s.

In addition to these basic parameters, several other important parameters are required to completely specify the external conditions to be used in wind turbine design. In the case of the wind turbine classes I_A through III_C, later referred to as the standard wind turbine classes, the values of these additional parameters are specified in 6.3, 6.4 and 6.5.

The design lifetime for wind turbine classes I to III shall be at least 20 years.

For the wind turbine class S the manufacturer shall, in the design documentation, describe the models used and values of design parameters. Where the models in Clause 6 are adopted, statement of the values of the parameters will be sufficient. The design documentation of wind turbine class S shall contain the information listed in Annex A.

The abbreviations added in parentheses in the subclause headings in the remainder of this clause are used for describing the wind conditions for the design load cases defined in 7.4.

6.3 Wind conditions

A wind turbine shall be designed to safely withstand the wind conditions defined by the selected wind turbine class.

The design values of the wind conditions shall be clearly specified in the design documentation.

The wind regime for load and safety considerations is divided into the normal wind conditions, which will occur frequently during normal operation of a wind turbine, and the extreme wind conditions that are defined as having a 1-year or 50-year recurrence period.

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1 The annual average wind speed no longer appears in Table 1 as a basic parameter for the wind turbine classes in this edition of the standard. The annual average wind speed for wind turbine designs according to these classes is given in equation (9).

2 Note that I_{ref} is defined as the mean value in this edition of the standard rather than as a representative value.

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The wind conditions include a constant mean flow combined, in many cases, with either a varying deterministic gust profile or with turbulence. In all cases, the influence of an inclination of the mean flow with respect to a horizontal plane of up to 8º shall be considered. This flow inclination angle shall be assumed to be invariant with height.

The expression "turbulence" denotes random variations in the wind velocity from 10 min. averages. The turbulence model, when used, shall include the effects of varying wind speed, shears and direction and allow rotational sampling through varying shears. The three vector components of the turbulent wind velocity are defined as:

– longitudinal – along the direction of the mean wind velocity;

– lateral – horizontal and normal to the longitudinal direction, and

– upward – normal to both the longitudinal and lateral directions, i.e. tilted from the vertical by the mean flow inclination angle.

For the standard wind turbine classes, the random wind velocity field for the turbulence models shall satisfy the following requirements:

a) the turbulence standard deviation, σ_1, with values given in the following subclauses, shall be assumed to be invariant with height. The components normal to the mean wind direction shall have the following minimum standard deviations[3]:

   – lateral component – $\sigma_2 \geq 0{,}7\sigma_1$

   – upward component – $\sigma_3 \geq 0{,}5\sigma_1$

b) the longitudinal turbulence scale parameter, Λ_1, at hub height z shall be given by

$$\Lambda_1 = \begin{cases} 0{,}7\,z & z \leq 60\,m \\ 42\,m & z \geq 60\,m \end{cases} \tag{5}$$

   The power spectral densities of the three orthogonal components, $S_1(f)$, $S_2(f)$, and $S_3(f)$ shall asymptotically approach the following forms as the frequency in the inertial sub-range increases:

$$S_1(f) = 0{,}05\,\sigma_1^2\,(\Lambda_1 / V_{\text{hub}})^{-2/3} f^{-5/3} \tag{6}$$

$$S_2(f) = S_3(f) = \tfrac{4}{3} S_1(f) \tag{7}$$

c) a recognized model for the coherence, defined as the magnitude of the co-spectrum divided by the auto-spectrum for the longitudinal velocity components at spatially separated points in a plane normal to the longitudinal direction, shall be used.

The recommended turbulence model that satisfies these requirements is the Mann uniform shear turbulence model in Annex B. Another frequently used model that satisfy these requirements is also given in Annex B. Other models should be used with caution, as the choice may affect the loads significantly.

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3 The actual values may depend on the choice of turbulence model and the requirements in b).

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6.3.1 Normal wind conditions

6.3.1.1 Wind speed distribution

The wind speed distribution is significant for wind turbine design because it determines the frequency of occurrence of individual load conditions for the normal design situations. The mean value of the wind speed over a time period of 10 min shall be assumed to follow a Rayleigh distribution at hub height given by

$$P_{\mathrm{R}}(V_{\mathrm{hub}}) = 1 - \exp\left[-\pi\left(V_{\mathrm{hub}} / 2V_{\mathrm{ave}}\right)^2\right] \tag{8}$$

where, in the standard wind turbine classes, V_{ave} shall be chosen as

$$V_{\mathrm{ave}} = 0,2\ V_{\mathrm{ref}} \tag{9}$$

6.3.1.2 The normal wind profile model (NWP)

The wind profile, $V(z)$, denotes the average wind speed as a function of height, z, above the ground. In the case of the standard wind turbine classes, the normal wind speed profile shall be given by the power law:

$$V(z) = V_{\mathrm{hub}}\left(z / z_{\mathrm{hub}}\right)^{\alpha} \tag{10}$$

The power law exponent, α, shall be assumed to be 0,2.

The assumed wind profile is used to define the average vertical wind shear across the rotor swept area.

6.3.1.3 Normal turbulence model (NTM)

For the normal turbulence model, the representative value of the turbulence standard deviation, σ_1, shall be given by the 90 % quantile[4] for the given hub height wind speed. This value for the standard wind turbine classes shall be given by

$$\sigma_1 = I_{\mathrm{ref}}\left(0,75V_{\mathrm{hub}} + b\right); \quad b = 5,6\ \mathrm{m/s} \tag{11}$$

Values for the turbulence standard deviation σ_1 and the turbulence intensity $\sigma_1/V_{\mathrm{hub}}$ are shown in Figures 1a and 1b.

Values for I_{ref} are given in Table 1.

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4 Note, if other quantiles are desired for additional optional load calculations, they may be approximated for the standard classes by assuming a log-normal distribution and

$E\langle\sigma_1 | V_{\mathrm{hub}}\rangle = I_{\mathrm{ref}}\left(0,75V_{\mathrm{hub}} + c\right);\ c = 3,8\ \mathrm{m/s}$

$Var\langle\sigma_1 | V_{\mathrm{hub}}\rangle = \left(I_{\mathrm{ref}}\left(1,4\ \mathrm{m/s}\right)\right)^2$

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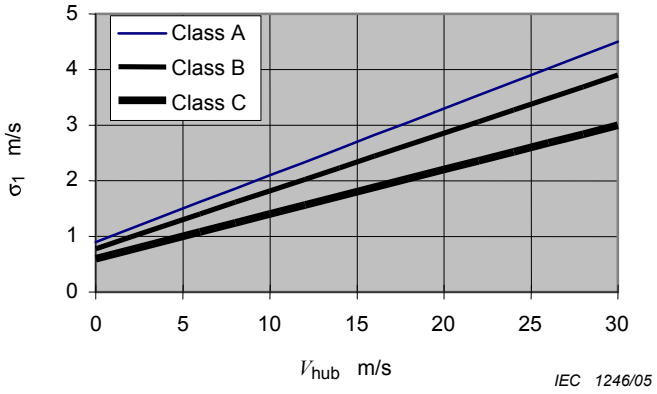


Figure 1a –Turbulence standard deviation for the normal turbulence model (NTM)

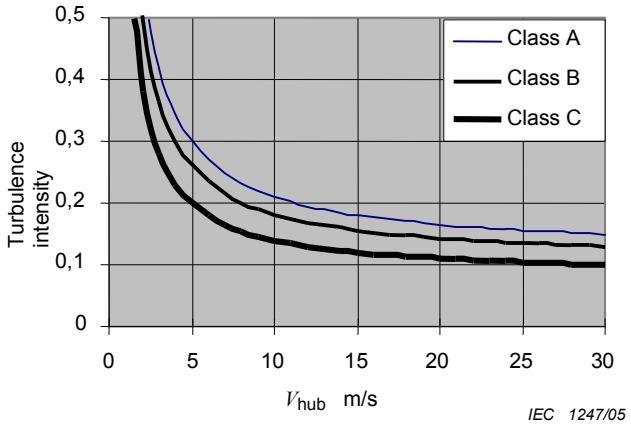


Figure 1b – Turbulence intensity for the normal turbulence model (NTM)

Figure 1 – Normal turbulence model (NTM)

6.3.2 Extreme wind conditions

The extreme wind conditions include wind shear events, as well as peak wind speeds due to storms and rapid changes in wind speed and direction.

6.3.2.1 Extreme wind speed model (EWM)

The EWM shall be either a steady or a turbulent wind model. The wind models shall be based on the reference wind speed, V_{ref}, and a fixed turbulence standard deviation, σ_1.

For the steady extreme wind model, the extreme wind speed, V_{e50}, with a recurrence period of 50 years, and the extreme wind speed, V_{e1}, with a recurrence period of 1 year, shall be computed as a function of height, z, using the following equations:

$$V_{\text{e50}}(z) = 1{,}4 V_{\text{ref}} \left(z / z_{\text{hub}} \right)^{0{,}11} \tag{12}$$

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B.2 Kaimal (1972)[19] spectrum and exponential coherence model

The component power spectral densities are given in non-dimensional form by the equation:

$$\frac{f\,S_k(f)}{\sigma_k^2} = \frac{4 f L_k / V_{hub}}{\left(1 + 6 f L_k / V_{hub}\right)^{5/3}} \qquad \text{(B.14)}$$

where

f is the frequency in Hertz,

k is the index referring to the velocity component direction (i.e. 1 = longitudinal, 2 = lateral, and 3 = upward);

S_k is the single-sided velocity component spectrum;

σ_k is the velocity component standard deviation (see equation (B.2));

L_k is the velocity component integral scale parameter,

and with

$$\sigma_k^2 = \int_0^\infty S_k(f)\,df \qquad \text{(B.15)}$$

The turbulence spectral parameters are given in Table B.1.

Table B.1 – Turbulence spectral parameters for the Kaimal model

	Velocity component index (*k*)		
	1	2	3
Standard deviation σ_k	σ_1	0,8 σ_1	0,5 σ_1
Integral scale, L_k	8,1 Λ_1	2,7 Λ_1	0,66 Λ_1

where σ_1 and Λ_1 are the standard deviation and scale parameters, respectively, of the turbulence as specified in 6.3.

The following exponential coherence model may be used in conjunction with the Kaimal autospectrum to account for the spatial correlation structure of the longitudinal velocity component:

$$\mathrm{Coh}(r,f) = \exp\left[-12\left(\left(f \cdot r / V_{hub}\right)^2 + \left(0{,}12\, r / L_c\right)^2\right)^{0{,}5}\right] \qquad \text{(B.16)}$$

where

Coh(r,f) is the coherence function defined by the complex magnitude of the cross-spectral density of the longitudinal wind velocity components at two spatially separated points divided by the autospectrum function;

r is the magnitude of the projection of the separation vector between the two points on to a plane normal to the average wind direction;

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19 Note that the turbulence component variance ratios in Table B.1 and the equation form for the upward velocity component differ somewhat from the original Kaimal spectral model. The longitudinal scale has been chosen to approximate the original Kaimal spectrum and, for the lateral and upward scales, to satisfy the spectral requirements in 6.3 for the asymptotic inertial subrange and the variance ratios given in Table B.1.

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f is the frequency in Hertz;

$L_{c} = 8{,}1\Lambda_1$ is the coherence scale parameter.

B.3 Reference documents

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